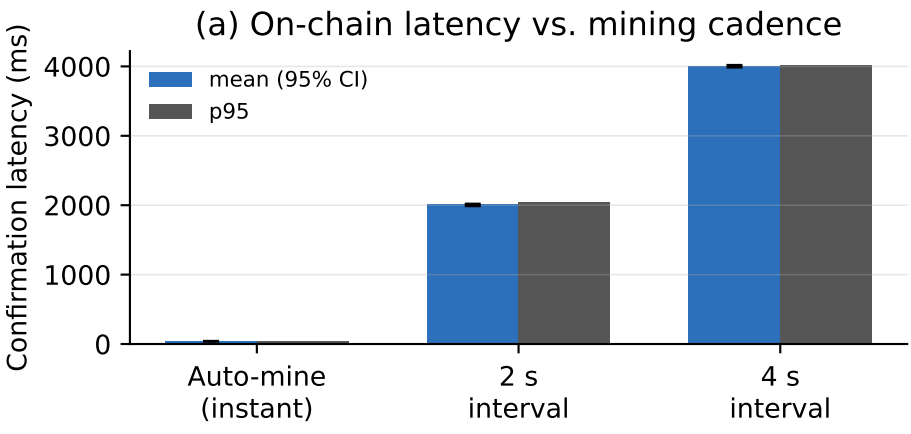
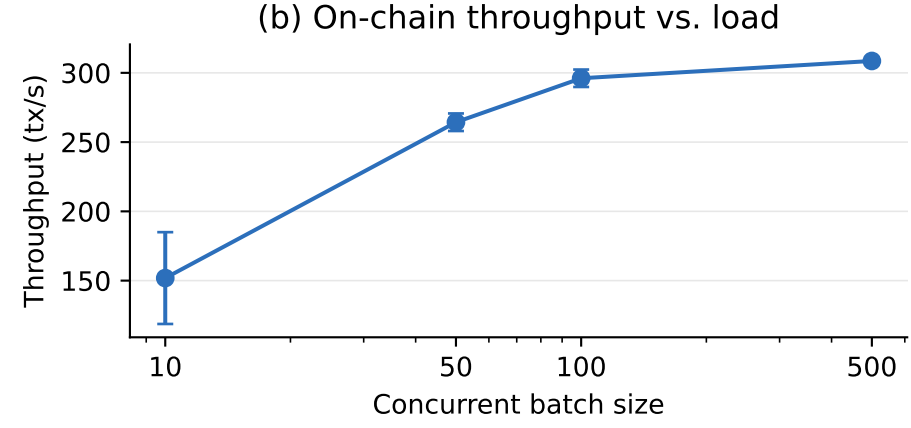
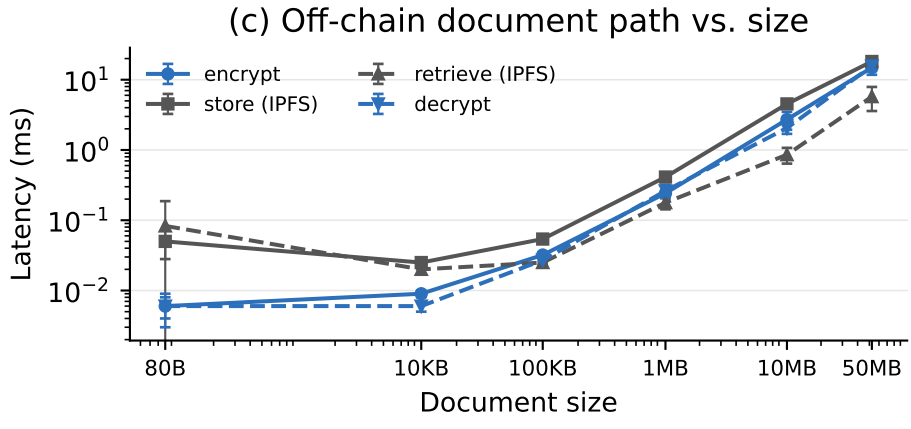
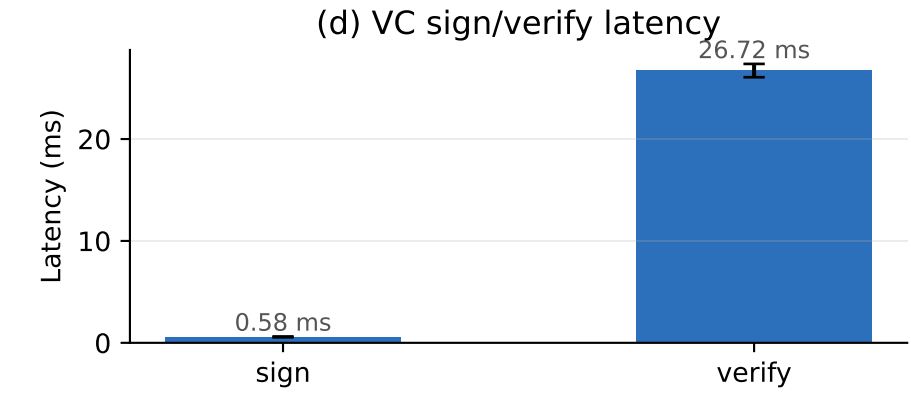

(a) On-chain latency vs. mining cadence
Confirmation latency (ms)
4000
3000
2000
1000
0
mean (95% CI)
p95
Auto-mine
(instant)
2 s
interval
4 s
interval


(b) On-chain throughput vs. load
Throughput (tx/s)
300
250
200
150
10
50
100
500
Concurrent batch size


(c) Off-chain document path vs. size
Latency (ms)
encrypt
store (IPFS)
retrieve (IPFS)
decrypt
80B
10KB
100KB
1MB
10MB
50MB
Document size


(d) VC sign/verify latency
Latency (ms)
26.72 ms
0.58 ms
20
10
0
sign
verify